SIMATS ENGINEERING
Department of Computer Science & Engineering
ASSESSMENT TOOL 2- MCQ TEST

Course Code:	**CSA12**	**Course Title:**	**Computer Architecture**
CO Assessed:	**CO3**	**Assessment:**	**ASSESSMENT TOOL 2- MCQ TEST**
Weightage:	**25%**	**Total Marks:**	**25**
CO3 Statement:	*Design processor organization by constructing pipelined datapath structures, identifying hazard types (data, control, structural), and comparing superscalar techniques such as out-of-order execution, speculative execution, and simultaneous multithreading (SMT). (BL6)*		
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Department:	**BE CSE**	**Date:**	

ANNEXURE A

1. A processor transfers data between registers before performing an arithmetic operation in the ALU. Which component primarily controls these register transfer operations?
   a) Arithmetic Logic Unit
   b) Control Unit
   c) Memory Unit
   d) Cache Controller

Answer: b) Control Unit

2. In a system where multiple registers are used for intermediate storage, data must be efficiently moved before computation. Which concept ensures correct sequencing of such operations?
   a) Register transfer language
   b) Instruction pipelining
   c) Cache mapping
   d) Interrupt handling

Answer: a) Register transfer language

3. A processor performs logical AND, OR, and XOR operations on data stored in registers. These operations are handled by which unit?
   a) Control Unit
   b) ALU
   c) Bus Interface Unit
   d) Memory Management Unit

Answer: b) ALU

4. A system designer chooses a single-bus architecture for simplicity, but performance issues arise when multiple components attempt data transfer simultaneously. What is the primary limitation here?
   a) Lack of memory
   b) Bus contention
   c) High clock speed
   d) Complex control logic

Answer: b) Bus contention

5. In a multi-bus organization, data transfer between CPU and memory is faster compared to a single-bus system. What is the main reason for this improvement?
   a) Reduced instruction size

b) Parallel data transfer capability
c) Increased clock frequency
d) Simplified hardware

Answer: b) Parallel data transfer capability

6. A processor executes instructions using a 5-stage pipeline. If one stage takes significantly longer than others, what will be the impact?
   a) Increased throughput
   b) Pipeline stalls
   c) Reduced instruction size
   d) Faster execution

Answer: b) Pipeline stalls

7. During pipelined execution, an instruction depends on the result of a previous instruction that has not yet completed. What type of hazard does this represent?
   a) Structural hazard
   b) Control hazard
   c) Data hazard
   d) Memory hazard

Answer: c) Data hazard

8. A pipeline encounters delays due to branching instructions, causing incorrect instruction fetching. Which hazard is this?
   a) Data hazard
   b) Control hazard
   c) Structural hazard
   d) Timing hazard

Answer: b) Control hazard

9. In a system where multiple instructions require the same hardware resource simultaneously, execution is delayed. This is an example of:
   a) Data hazard
   b) Structural hazard
   c) Control hazard
   d) Logical hazard

Answer: b) Structural hazard

10. To reduce pipeline stalls caused by data dependencies, a processor implements forwarding. What does forwarding achieve?
    a) Reduces instruction size
    b) Bypasses intermediate storage
    c) Increases clock speed
    d) Eliminates memory access

Answer: b) Bypasses intermediate storage

11. A superscalar processor issues multiple instructions per clock cycle. What enables this capability?
    a) Single execution unit
    b) Multiple execution units
    c) Larger memory
    d) Slower clock

Answer: b) Multiple execution units

12. In out-of-order execution, instructions are executed in a different order than they appear in the program. What is the main advantage of this technique?
    a) Reduced memory usage
    b) Better utilization of CPU resources
    c) Simplified instruction set
    d) Reduced instruction count

Answer: b) Better utilization of CPU resources

13. A processor predicts the outcome of a branch instruction and executes instructions ahead of time. This technique is known as:

a) Pipelining
b) Speculative execution
c) Multiprocessing
d) Interrupt handling

Answer: b) Speculative execution

14. In speculative execution, incorrect predictions may lead to discarded results. What is the main drawback of this approach?

a) Increased memory size
b) Wasted computation
c) Reduced clock speed
d) Increased instruction size

Answer: b) Wasted computation

15. A processor uses separate units to dispatch instructions to execution units. What is the primary role of the dispatch unit?

a) Execute instructions
b) Fetch instructions
c) Assign instructions to execution units
d) Store instructions

Answer: c) Assign instructions to execution units

16. Execution units in a processor handle different types of operations simultaneously. Which of the following is an example of an execution unit?

a) Cache memory
b) ALU
c) Control Unit
d) Register file

Answer: b) ALU

17. A processor with hyper-threading allows multiple threads to share execution resources. What is the primary benefit?

a) Reduced hardware cost
b) Increased instruction throughput
c) Lower power consumption
d) Smaller memory

Answer: b) Increased instruction throughput

18. In simultaneous multithreading (SMT), multiple threads are executed at the same time. What challenge arises from this approach?

a) Reduced instruction set
b) Resource contention
c) Lower clock speed
d) Reduced memory

Answer: b) Resource contention

19. A system experiences reduced performance when multiple threads compete for the same execution units. This issue is related to:

a) Instruction pipelining
b) Resource sharing
c) Cache coherence
d) Memory hierarchy

Answer: b) Resource sharing

20. A processor pipeline achieves ideal performance when all stages take equal time. What happens if stages are unbalanced?

a) Improved speed
b) Reduced efficiency
c) Increased memory
d) No effect

Answer: b) Reduced efficiency

21. In a multi-bus system, data can be transferred between different components simultaneously. What is the main advantage of this design?

a) Simplicity
b) Parallelism
c) Lower cost
d) Reduced memory

Answer: b) Parallelism

22. A processor encounters frequent stalls due to instruction dependencies. Which technique can best improve performance?

a) Increasing memory size
b) Data forwarding
c) Reducing clock speed
d) Decreasing instruction count

Answer: b) Data forwarding

23. A superscalar processor executes instructions out of order but commits them in order. Why is this necessary?

a) To reduce memory usage
b) To maintain program correctness
c) To increase clock speed
d) To simplify hardware

Answer: b) To maintain program correctness

24. A processor uses speculative execution along with branch prediction. What determines the effectiveness of this approach?

a) Memory size
b) Prediction accuracy
c) Instruction size
d) Number of registers

Answer: b) Prediction accuracy

25. A computing system integrates pipelining, superscalar execution, and SMT. What is the overall goal of combining these techniques?

a) Reduce hardware complexity
b) Increase instruction throughput
c) Decrease memory usage
d) Simplify programming

Answer: b) Increase instruction throughput

1. **Answer:** b) Control Unit
2. **Answer:** a) Register transfer language
3. **Answer:** b) ALU
4. **Answer:** b) Bus contention
5. **Answer:** b) Parallel data transfer capability
6. **Answer:** b) Pipeline stalls
7. **Answer:** c) Data hazard
8. **Answer:** b) Control hazard
9. **Answer:** b) Structural hazard
10. **Answer:** b) Bypasses intermediate storage
11. **Answer:** b) Multiple execution units
12. **Answer:** b) Better utilization of CPU resources
13. **Answer:** b) Speculative execution
14. **Answer:** b) Wasted computation
15. **Answer:** c) Assign instructions to execution units
16. **Answer:** b) ALU
17. **Answer:** b) Increased instruction throughput
18. **Answer:** b) Resource contention
19. **Answer:** b) Resource sharing
20. **Answer:** b) Reduced efficiency
21. **Answer:** b) Parallelism
22. **Answer:** b) Data forwarding
23. **Answer:** b) To maintain program correctness
24. **Answer:** b) Prediction accuracy
25. **Answer:** b) Increase instruction throughput

Code of Conduct:

I G.SARANRAJ / 192511182 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

MARKS ALLOCATION

	Understandin Concepts (25	Accuracy of An (30%)	Application Knowledge (	Time Manage (15%)	Logical Thinkin Decision Making
Q1					
Q2					
Q3					
Q4					
Q5					

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginn
Understanding of Concepts	**25%**	Demonstrates clear and comp understanding of concepts tes MCQs	Shows good understa with minor conceptu	Partial understanding; key concepts	Misinterprets or f understand conce
Accuracy of Answ	**30%**	Selects correct answers for al questions with high accuracy	Answers most questi correctly with few er	Provides limited correc answers with frequent	Unable to select c answers consiste
Application of Kn	**20%**	Applies concepts effectively t application-based and analyti questions	Applies concepts wit errors	Limited application of concepts; requires improvement	Unable to apply r knowledge
Time Managemen	**15%**	Completes all questions withi allotted time effectively	Completes most ques within time	Attempts questions but struggles with time management	Unable to comple test on time
Logical Thinking Decision Making	**10%**	Analyzes options carefully an chooses the most appropriate	Makes correct choice reasonable analysis	Shows limited reasoni selecting answers	Answers without reasoning